Heterogenous price patterns of cryptocurrency between exchanges
: Pre-averaging realized volatility matrix and its analysis

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MEN891M Final Report (2025.12.14.) Team 1

Inha Jeong
Jeongwoon Kim
Seungjun Oh

1. Introduction

Cryptocurrency is a virtual currency distributed by block chain that records trading history through various computers simultaneously. There are no monopoly institutions that control the volume and trading history. Due to this property, price of cryptocurrency can be different among the exchanges. From this, there are some traders who try to catch the gap and get the arbitrage. Kimchi premium is a term describing this price gap between the exchanges. The heterogenous prices between the exchanges imply different risks of the currency. For example, liquidity or availability to trade financial assets without large changes in the price can be different between the exchanges due to the number of participants trading the cryptocurrency (European Central Bank, 2007). This indicate that property of a certain cryptocurrency can be different by the exchanges.

From this fact, this research tries to analyze heterogenous patterns of price and relationships between exchanges. In this study, first, we do martingale test to see existence of market efficiency in the cryptocurrency. The market efficiency means that prediction of return is not possible since current prices reflect all history or past information. And a stochastic process $\{x_t\}_{t\geq 0}$ filtration $\{F_s\}$ with satisfies martingale property if $E(F_s) = x_s$ and time $s \leq t$. Since martingale property guarantees no extra term in the conditional expectation given historical data, which is consistent with the market efficiency. This is one of important assumptions for portfolio theory, we check the property in the cryptocurrency.

Next, to check robustness of the martingale test results and measure predictability of price dynamics, we estimate Hurst exponent and set heterogenous autoregression (HAR) model. Hurst exponent is to control smoothness of fractional Brownian motion. If the value is 0.5, then its corresponding process is Brownian motion having martingale test. If value is not 0.5, then martingale property is not hold, implying possibility to predict conditional expectation given historical data. Since the Hurst exponent does not show model, HAR model is used to predict future values.

This paper will proceed as follows. Section 2 is about data we use for analysis and Section 3 is about concrete explanation of methodology. Section 4 shows results and interpretation, and section 5 is to conclude our study.

2. Data

This chapter details the data collection strategy and preprocessing steps as a prerequisite for analyzing the efficiency and regional premium of the global cryptocurrency market. A precise time-series analyzable dataset was constructed by collecting high-frequency (High-Frequency) data from major exchanges representing each region and standardizing it by converting it to a unified base currency.

2.1. Data Gathering

2.1.1. Collection Scope and Target

The data collection targets for this study encompass trading pairs between Bitcoin (BTC) and fiat currencies, and between the stablecoin Tether (USDT) and fiat currencies, listed on major global cryptocurrency exchanges. The analysis period was set to the most recent one year, from November 2024 to October 2025, to reflect the

latest market trends. Specifically, the data collection scope was set to high-resolution data at 1-second and 1-minute intervals, going beyond typical minute-level data, to enable High-Frequency market data's microstructure analysis.

2.1.2. Exchange Selection

To reflect the regional characteristics of the global cryptocurrency market and the features of each exchange, data collection was attempted by selecting six well-known representative exchanges in the industry (Binance, Bitstamp, Bybit, Gemini, OKX, Upbit). For regional characteristics, data for BTC pairs based on the fiat currency of each continent and fiat/USDT pairs that can replace the exchange rate were collected. In the initial data collection stage, various currency pairs from major exchanges were secured as Raw Data for comprehensive analysis (see Table 1). BTC/USDT was collected as a global indicator used worldwide, followed by BTC pair data based on fiat currencies from Europe (EUR, GBR), South America (BRZ, BRL), and Asia (KRW, TRY, SGD).

For the major exchanges such as Binance, Bybit, OKX, and Upbit, data were collected directly through the official APIs and Historical Data Collection Pages provided by each exchange. For some data where official data acquisition was limited (Bitstamp, Gemini, etc.), data was secured through a reliable third-party data provider (CryptoDataDownload) to enable cross-validation.

2.1.3. Data Filtering

Data lacking liquidity or not aligning with the research objectives among the collected data can compromise the reliability of the analysis results. Therefore, the following strict criteria were applied for filtering (see table 1 for filtered pairs):

1. Exclusion of Insufficient Liquidity: Illiquid trading pairs with an average trading frequency of less than once every 5 minutes were excluded, as they were judged to lack statistical significance.
2. Exclusion of Incomplete Proxy Pairs: For the core methodology of this study, 'Proxy Generation within the Same Exchange, pairs where a BTC/Fiat pair and a USDT/Fiat pair did not simultaneously exist within a specific exchange were excluded. That is, data that could not form a pair and thus could not calculate the dollar-equivalent price were not included in the final dataset to maintain consistency.

Exchange	Raw Data Pairs	Filtered Data Pairs
Binance	BTC/USDT, BTC/BRL, BTC/EUR, EUR/USDT, USDT/BRL	BTC/USDT, BTC/BRL, BTC/EUR, EUR/USDT, USDT/BRL
Bitstamp	BTC/EUR, BTC/GBP, GBP/USDT, USDT/EUR	-
Bybit	BTC/BRZ, BTC/EUR, BTC/TRY, USDT/BRZ, USDT/EUR, USDT/TRY	BTC/EUR, USDT/EUR
Gemini	BTC/EUR, BTC/GBP, BTC/SGD, BTC/USDT	-

OKX	BTC/BRL, BTC/EUR, BTC/USDT, USDT/BRL, USDT/EUR	BTC/BRL, BTC/EUR, USDT/BRL, USDT/EUR
Upbit	BTC/KRW, USDT/KRW	BTC/KRW, USDT/KRW

Table 1: List of currency pairs for each exchange

2.2. Data Preprocessing

The collected Raw Data lacked uniformity in format, such as tick data or candle data. Furthermore, using tick data directly could lead to interpretation errors when comparing unsynchronized data. Therefore, the following stepwise preprocessing was performed to convert it into time-series data suitable for comparative analysis.

2.2.1. OHLCV Data Construction

Many global exchanges often do not directly provide sub-minute candle data due to server load, etc. To overcome this limitation, this study adopted the method of collecting individual transaction details (Tick data) when 1s-unit candle data was not provided, and then directly reconstructing 1-second and 1-minute OHLCV (Open, High, Low, Close, Volume) data based on it. In this process, the Open Price for each time interval was defined as the price executed at the exact Unix timestamp when the interval began, maintaining data consistency.

2.2.2. Proxy Data Generation

Direct price comparison is impossible because regional exchanges trade Bitcoin with different fiat currencies such as Korean Won (KRW), Euro (EUR), and Real (BRL). To solve this, this study applied the methodology of generating a 'BTC/USDT Proxy Pair.' The proxy price is calculated by product of two matching pairs (BTC/Fiat * Fiat/USDT). The most important prerequisite here was to only combine pairs traded 'within the Same Exchange' for the calculation. This was done to minimize the dilution of exchange information during the proxy generation process.

2.3. Data Collection Results

The final dataset constructed through the collection and preprocessing steps described above encompasses major economic zones including Global, Europe, South America, and East Asia, with the specific composition as follows.

Data composition by Exchange

- Binance: BTC/BRL, BTC/EUR, BTC/USDT, EUR/USDT, USDT/BRL
- Bybit: BTC/EUR, USDT/EUR
- OKX: BTC/BRL, BTC/EUR, USDT/BRL, USDT/EUR
- Upbit: BTC/KRW, USDT/KRW

Data Composition by Regional

- Global (USDT): Binance
- Europe (EUR): Binance, Bybit, OKX
- South America (BRL): Binance, OKX
- East Asia (KRW): Upbit

3. Method

3.1. Martingale test

Martingale test is done to see existence of information efficiency in the cryptocurrency or its volatility. It checks availability to predict future data using history. Using the fact $\Delta y_t = y_t - y_{t-1},\ E(\Delta y_t | y_{t-1}) = 0\ a.s\ iff\ E\big(\Delta y_t I\ (y_{t-1} \le x)\big) = 0\ for\ a.s\ x \in\ R$, the following test statistics S_n and T_n for the timeseries $\{y_t\}_{t=1}^n$ are used for the test (Park and Whang, 2004):

$$Q_n(x) = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} \Delta y_t I(y_{t-1} \le x)\,,\ S_n = \sup_{x \in R} |Q_n(x)|,\ T_n = \frac{1}{n} \sum_{t=1}^{n} Q_n^2(y_{t-1})$$

The $Q_n(x)$ is a statistic for $E\big(\Delta y_t I\ (y_{t-1} \le x)\big)$. The Kolmogorov-Smirnov statistic S_n and Cramer-von Mises type statistic T_n is used due to their consistency if some conditions are satisfied. The consistency provides some appropriate distribution for the test, so we do the test using the statistics. Instead of using Filtration $F_t = \{y_s\}_{s=1}^t$, Park and Whang (2004) use Filtration $F_{t-1} = \{y_{t-1}\}$ to simplify test. This may make bias by not considering all previous data or delay, so we try to supplement the test with Hurst exponent or HAR.

3.2. Estimation of Hurst exponent

To supplement Martingale test result and measure information delay in the volatility, we also estimate Hurst exponent of fractional Brownian motion to measure information delay or other patterns. Since many data are expressed as the conditional normal distribution given some deterministic terms, the fractional Brownian motion is appropriate in many cases. A fractional Brownian motion $\{B_t^H\}$ is defined as $B_t^H = \frac{1}{\Gamma(H+0.5)} \int_0^t (t-s)^{H-0.5} dB_s$, which is generalized version of derivatives by allowing fractional derivatives.

If $H > 0.5$, it implies that B_t^H is smoother by allowing derivatives with respect to time t (derivates of Brownian motion can't be defined). Since smooth function has small derivatives that preserve more global information, big H can preserve geometric structure and reduce jumps. If $H < 0.5$, it implies negative derivative which means that there are some patterns (e.g., previous increment results in future decrement). For data $\widehat{B_k} = B_k - E(B_k)$, the estimation of H is based on the following equations (Lo, 1991):

$$\frac{R(n)}{S(n)} = Cn^H\ for\ Z_i = \sum_{k=1}^{i} \left(\widehat{B_k}\right),\ R(n) = \max_i(Z_i) - \min_i(Z_i)\ \ and\ S(n) = \sqrt{\frac{1}{n} \sum_{k=1}^{n} \widehat{B_k}^2}$$

The H is estimated by regression for $log\frac{R(n)}{S(n)} = logR(n) - \log S(n) = logC + hlog(n)$. Since Brownian motion has martingale property and its H is 0.5, statistical test for hypothesis H_0: $H = 0.5$ is done by Monte Carlo simulation for motion H of Brownian motion.

3.3. Pre-averaging Realized Volatility Matrix (PRVM)

To quantify the volatility dynamics of multiple assets, we employ the pre-averaging realized volatility matrix (PRVM), which is a generalized multivariate extension of the pre-averaging realized volatility estimator. Unlike the conventional realized covariance

matrix constructed from high-frequency returns, the PRVM is specifically designed to reduce the adverse effects of market microstructure noise, which tends to dominate volatility estimates at very high sampling frequencies. The PRVM is formally defined as follows:

$$\Gamma_{d,\,ij} = \frac{1}{\psi K} \sum_{k=1}^{n-K+1} \left\{ \bar{Y}_i(t_{d,k}) \bar{Y}_j(t_{d,k}) - \frac{1}{2} \hat{Y}_{ij}(t_{d,k}) \right\},$$

$where$

$\bar{Y}_i(t_{d,k}) = \sum_{l=1}^{K-1} g\left(\frac{l}{K}\right) \left(Y_i(t_{d,k+l}) - Y_j(t_{d,k+l-1}) \right)$,

$\hat{Y}_{ij}(t_{d,k}) = \sum_{l=1}^{K} \left[\left\{ g\left(\frac{l}{K}\right) - g\left(\frac{l-1}{K}\right) \right\}^2 \left(Y_i(t_{d,k+l-1}) - Y_i(t_{d,k+l-2}) \right) \left(Y_j(t_{d,k+l-1}) - Y_j(t_{d,k+l-2}) \right) \right]$.

3.4. Matrix Heterogenous Autoregressive Regression (MHAR)

To capture heterogeneous dynamics and interconnections across different exchanges and currencies, we extend the standard autoregressive (AR) model for volatility series. Specifically, we incorporate the pre-averaging realized volatility matrix (PRVM), which quantifies both the cross-exchange and cross-currency relationships as well as individual asset volatility. We refer to this specification as the matrix autoregressive (MAR) model. To improve model complexity and account for heterogeneity across different time interval lengths, we embedded the heterogeneous autoregressive (HAR) structure into the MAR framework. This extension results in the matrix heterogeneous autoregressive (MHAR) model. The definitions of each model are provided below:

$$\hat{\sigma}_t^2 = c + \beta_1 \sigma_{t-1}^2 + \beta_2 \sigma_{t-2}^2 + \cdots + \beta_h \sigma_{t-h}^2 \text{ (AR)}$$

$$\hat{\sigma}_t^2 = c + \boldsymbol{\beta_1}^\top vech(\boldsymbol{\Gamma_{t-1}}) + \boldsymbol{\beta_2}^\top vech(\boldsymbol{\Gamma_{t-2}}) + \cdots + \boldsymbol{\beta_h}^\top vech(\boldsymbol{\Gamma_{t-h}}) \text{ (MAR)}$$

$$\hat{\sigma}_t^2 = c + \beta_{hour} \sigma_{t-1}^2 + \beta_{half-day} \frac{1}{12} \sum_{i=1}^{12} \sigma_{t-i}^2 + \beta_{day} \frac{1}{24} \sum_{i=1}^{24} \sigma_{t-i}^2 \text{ (HAR)}$$

$$\hat{\sigma}_t^2 = c + \boldsymbol{\beta_{hour}}^\top vech(\boldsymbol{\Gamma_{t-1}}) + \boldsymbol{\beta_{half-day}}^\top \frac{1}{12} \sum_{i=1}^{12} vech(\boldsymbol{\Gamma_{t-i}})$$
$$+ \boldsymbol{\beta_{day}}^\top \frac{1}{24} \sum_{i=1}^{24} vech(\boldsymbol{\Gamma_{t-i}}) \text{ (MHAR)}$$

4. Result

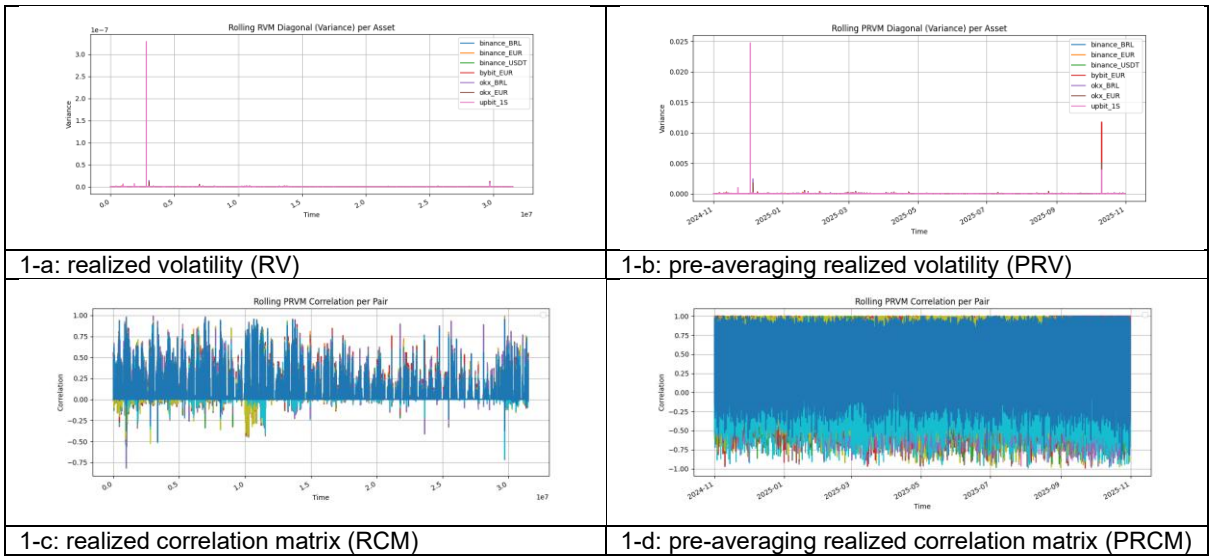
4.1.1. Comparison between Realized volatility and pre-averaging realized matrix

Not only Values of pre-averaging realized volatility (PRV) are much higher than Realized volatility (RV), but also spike patterns are revealed in the PRV. Also, in the pre-averaging realized correlation matrix (PRCM), negative correlations are dominated, which can't be observed in the realized correlation matrix (RCM). These results show analysis for the different exchanges can have bias if the micro-structure noise isn't be reduced properly.

One of variables that causes the noise is bid-ask spread or gap between prices referred by sellers and buyers (Clinet and Patiron, 2019). Since this gap is the difference in willingness to pay between people, this can be related to uncertainty of price or information asymmetry. Another variable that makes up the micro-structure noise is the information delay in the price (Li et al., 2020), which is caused by block-chain mechanism. Since history of bitcoin trading are recorded by another participants

and the speed depends on the network traffic or commission, prices may not reflect information immediately. From these facts, PRV and PRCM can reduce bias caused by other exogenous factors.

We can see negative correlations between some exchanges. This implies two things. First, there is arbitrage opportunity by buying cheaper cryptocurrency and selling it to other exchange. This different price patterns for the same cryptocurrency are caused by decentralization and information delay. For example, if a price of cryptocurrency decreases in a certain exchange but other exchange doesn't show the decrement yet, then negative correlation can be shown. Second, using the correlation between exchanges and min-variance portfolio strategy, we can diversify investments in the same cryptocurrency. The strategy distinguish cryptocurrency by exchanges, even for the same assets.



1-a: realized volatility (RV) | 1-b: pre-averaging realized volatility (PRV)

1-c: realized correlation matrix (RCM) | 1-d: pre-averaging realized correlation matrix (PRCM)

Figure 1: Realized volatility and pre-averaging realized volatility or covariance matrix (1s)

4.1.2. Martingale test results and Hurst exponent

Component	Sn (RVM)	Tn (RVM)	Sn (PRVM)	Tn (PRVM)
Var (Binance, BRL)	1.70E-07	1.58E-14	1.70E-07	1.58E-14
Var (Binance, EUR)	1.24E-07	1.57E-14	1.24E-07	1.57E-14
Var (Binance, USDT)	1.44E-07	9.66E-15	1.44E-07	9.66E-15
Var (Bybit, EUR)	5.68E-07	4.44E-13	5.68E-07	4.44E-13
Var (Okx, BRL)	7.99E-07	2.26E-13	7.99E-07	2.26E-13
Var (Okx, EUR)	2.02E-07	3.52E-14	2.02E-07	3.52E-14
Var (Upbit)	2.85E-07	2.82E-14	2.85E-07	2.82E-14

Table 1: Results of Martingale test for variance (RVM, PRVM) traded in different exchanges and exchange rates

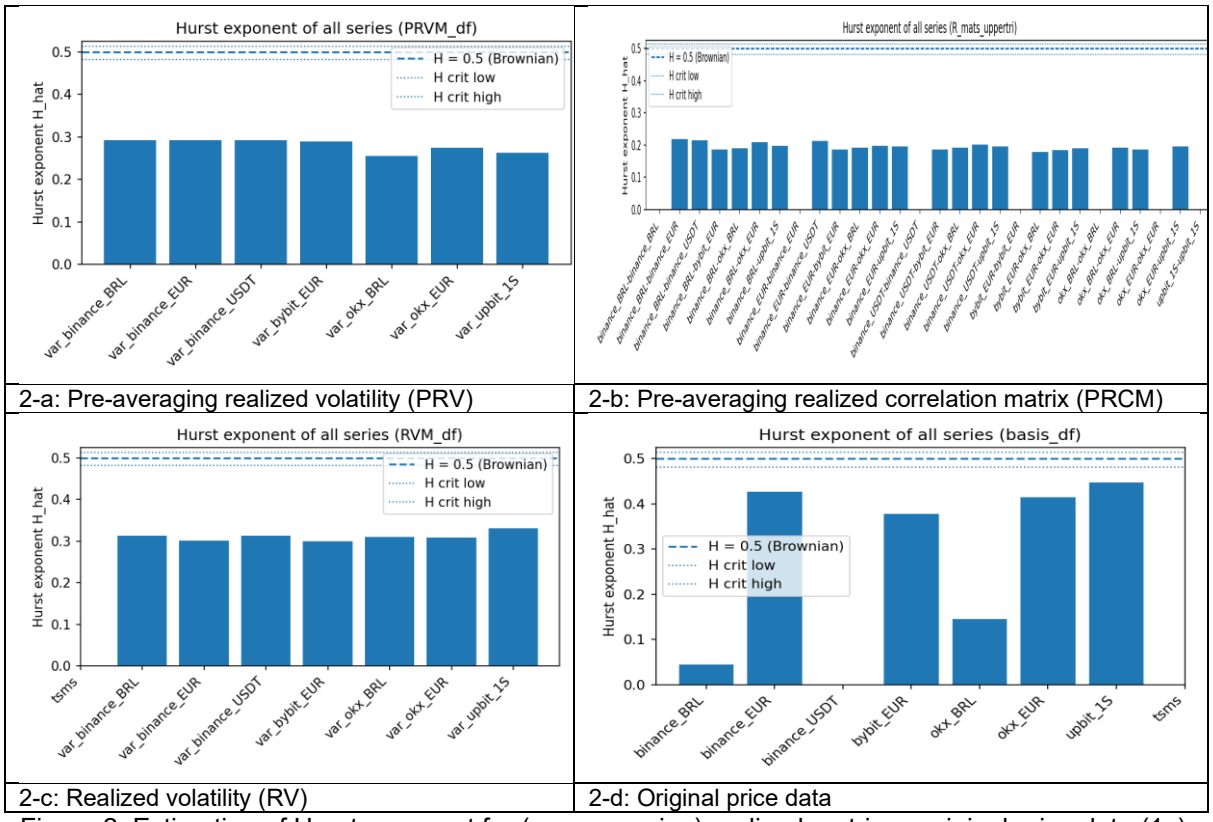
From the test, we can't reject Martingale property in the variance of Bitcoin (see Table 1). Since this test assumes $E(y_t|y_{t-1}) = y_{t-1}$, this test indicates that shocks in the variance persist overtime or random variables in the data have symmetry for the previous observation. This result is consistent with the results for 1m data. On the other hand, martingale tests are rejected except Bitcoin traded in the Binance with USD (see table 2). It implies that information efficiency only exists in the Bitcoin traded at Binance. The reason can be related to the facts that Binance is one of the largest trading exchanges and information are immediately reflected in the price by much trading. Another reason may be that Binance can process the trading immediately.

component	Sn	Tn	Reject to Sn=0	Reject to Tn=0
binance_BRL	5419.46888	18641145.5	TRUE	TRUE
binance_EUR	27.3874426	175.085418	TRUE	TRUE
binance_USDT	0	0	FALSE	FALSE
bybit_EUR	29.8882385	246.633943	TRUE	TRUE
okx_BRL	1779.5861	1658061.48	TRUE	TRUE
okx_EUR	25.893603	143.245963	TRUE	TRUE
upbit_1S	86984.6287	2262047050	TRUE	TRUE

Table 2: Results of Martingale test for original price data traded in different exchanges and exchange rates

As we mention before, since the Martingale test assume one previous data y_{t-1} as Filtration, the test can have bias. Also, it can't show property of dynamic pattern, we supplement the results by estimation of Hurst exponent. From Figure 2, all data have Hurst exponent smaller than 0.5. It means there are negative derivatives and fluctuation in the data. The negative feedback implies that the price will decrease if it is above the threshold and increase if it is below it (Heydari et al., 2014). The threshold can be true price of cryptocurrency. Due to the fluctuation or negative feedback, expectation of data tends to be constant in some period. This implies that if we know the threshold correctly, then we can predict movement of volatility or original data.

Since the Hurst exponent indicates possibility to predict tendency of the dynamics, this result may not be consistent with the Martingale test result that claims that there are no arbitrage from the prediction of future value using historical data. However, since negative Hurst exponent and martingale property imply existence of true information in the data, the two methods show consistent results.



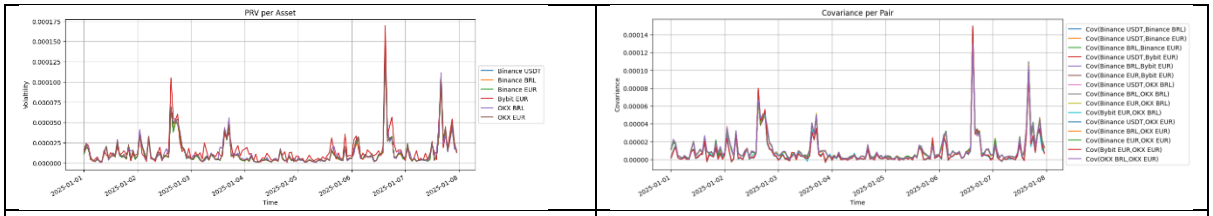
2-a: Pre-averaging realized volatility (PRV) | 2-b: Pre-averaging realized correlation matrix (PRCM)

2-c: Realized volatility (RV) | 2-d: Original price data

Figure 2: Estimation of Hurst exponent for (pre-averaging) realized matrix or original price data (1s)

4.1.3. Matrix Heterogenous Autoregression (MHAR)

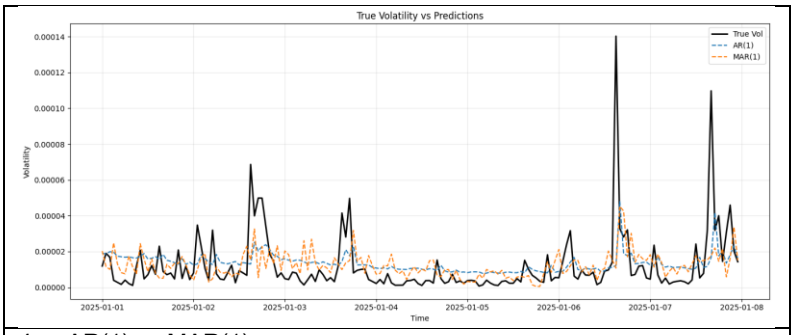
We compute the 1-hour PRVM using 1-minute BTC price data. Due to the extremely heavy-tailed behavior of the BTC/KRW series from Upbit, which we found to interfere with the stability of the regression analysis, this series is excluded from the construction of the volatility matrix. As a result, the PRVM is formed using 6 distinct exchange–currency pair proxies. The target time series for the AR-based volatility forecasting models is the pre-averaging volatility series of the most liquid asset, namely the 1-hour Binance BTC/USDT.



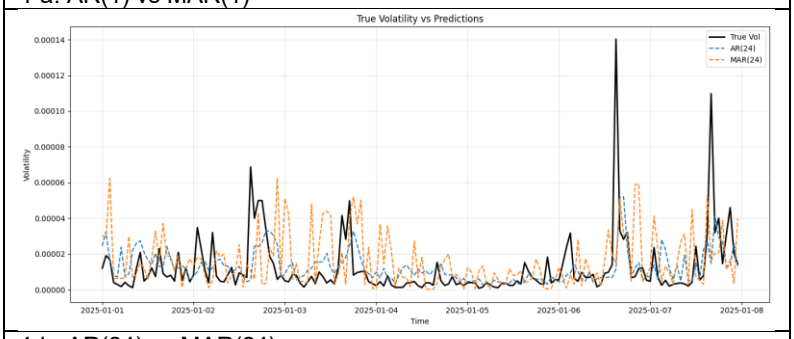
3-a: pre-averaging realized volatility (PRV) | 3-b: pre-averaging realized covariance

Figure 3: Pre-averaging realized matrix (1h)

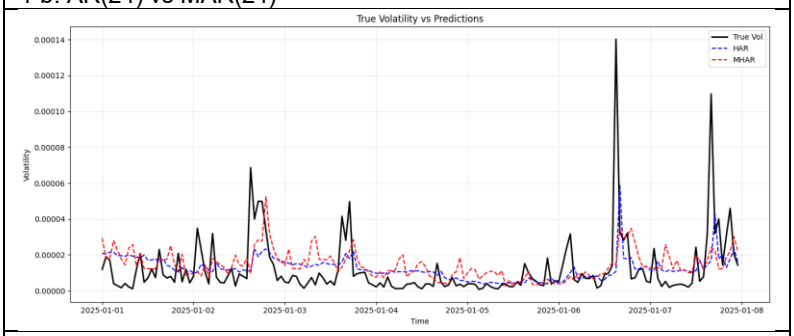
For the AR and MAR models, the time lags are set to 1 hour and 24 hours, and forecasts are generated accordingly. The forecasting results for the AR, MAR, HAR, and MHAR models are presented below.



4-a: AR(1) vs MAR(1)



4-b: AR(24) vs MAR(24)



4-c: HAR vs MHAR

Figure 4: Predicted and Realized Volatility time series

To evaluate the forecast performance of each model, we compute the mean squared error (MSE) and the QLIKE loss for all forecasts. The formal definitions of each loss function are provided below.

$$MSE(\hat{\sigma}^2) = \frac{1}{T}\sum_{t=1}^{T}(\hat{\sigma}_t^2 - \sigma_t^2)^2,$$

$$QLIKE(\hat{\sigma}^2) = \frac{1}{T}\sum_{t=1}^{T}\left[\log(\hat{\sigma}_t^2) + \frac{\sigma_t^2}{\hat{\sigma}_t^2}\right]$$

The results indicate that the MHAR model achieves the lowest MSE and a comparatively low QLIKE value.

	AR(24)	AR(1)	MAR(24)	MAR(1)	HAR	MHAR
$MSE(\times 10^7)$	2.360	1.015	1.116	1.044	1.055	0.8859
$QLIKE(\times 10^{-1})$	-1.015	-1.021	11.50	4.726	-1.020	-1.014

Table 3: MSE and QLIKE

5. Conclusion

This paper studies heterogenous patterns of cryptocurrency and relationships between currencies traded in the different exchanges. Due to micro-structured noise caused by some variables such as bid-ask spread or delay in the records, realized volatility covariance matrix has bias and pre-averaging can reduce it. And we observed several instances of negative correlations in the PRCM. This implies that even when holding the identical asset, risk management strategies that leverage the distinct characteristics of each exchange can be possible. Furthermore, this suggests the existence of idiosyncratic microstructures for the same cryptocurrency, varying by exchange and trading pair.

We confirmed that the variance of Bitcoin satisfies the martingale property. It implies that the shocks in the volatility persist over time. However, regarding Bitcoin prices, the martingale hypothesis was rejected in all cases with the sole exception of the Binance BTC/USDT pair. This can be interpreted as information efficiency is present only in the largest trading pair and there are excess returns using historical data in the other assets. However, it is necessary to interpret these results with the caveat that proxy data was utilized for pairs other than Binance BTC/USDT. idiosyncratic microstructures for the same cryptocurrency, varying by exchange and trading pair.

However, there are two limitations in this study. First, jump is not considered, so the pre-averaging matrix can have bias. Jump is that unexpected information is reflected in the prices. Due to decentralized property of cryptocurrency, there are some risks in the asset such as hacking, fraud or else. The effect of jump can be reduced by the thresholding method, but decision for the threshold may be subjective. Second, although we supplement the results of martingale test by the Hurst exponent and HAR model, martingale test is not done strictly because it doesn’t consider other history. And we don’t check existence of threshold in the volatility matrix to interpret negative Hurst exponent. For the interpretation, the threshold should be modelled to claim existence of true price.

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This study can be applied to risk management of cryptocurrency investments by considering heterogenous patterns of dynamics, relations between currencies traded in different exchanges or information delay.

6. Reference

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Team Member Roles and Contributions

Name	Major Responsibilities	Description of Contributions
Jeongwoon Kim	Martingale test, estimation of Hurst exponent, calculate pre-averaging matrix and realized matrix for 1s data	Data analysis for the 1s data
Inha Jeong	HAR model, calculate pre-averaging matrix and realized matrix for 1m data	Data analysis for the 1m data
Seungjun Oh	Data collection and pre-processing	Collect doens of gigabyte data and do pre-process for use